

# pibuilder Venture Strategy

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## | Findings, Analysis, and Roadmap

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## | Part 1 — The Honest Assessment (The Roast)

Before strategy, context. The following is a frank assessment of where things stood at the start of this conversation.

You have a hydrology platform where the only working module is rain. Fitting, because everything else about this project is also underwater.

Your SME — the one person who knows what half these Excel cells mean — is ghosting you in slow motion, and your strategy is apparently to hope the tension resolves itself while your ground truth walks out the door with their LinkedIn notification.

You've documented three bugs in an Excel file. Congratulations. You have a very well-tested reason to distrust a spreadsheet. That's ₹0 in ARR.

Your exit target is ₹830 crores. Your current asset is a CLI tool that tells you how much it rained near a mountain. The gap between those two numbers is not a roadmap, it's a hallucination.

That said — the code is clean, the architecture is sound, and the market is real. The rest of this document is the plan to close the gap.

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## | Part 2 — What Has Been Built

### Rainfall Frequency Analysis Engine (Phase 1 — Complete)

The only fully implemented module. It automates the rainfall frequency analysis currently done manually in Excel across EPC hydrology projects.

Two workflows:

| Workflow                | Data source                                                        | Output                                 |
|-------------------------|--------------------------------------------------------------------|----------------------------------------|
| Grid-based (MPSB)       | Satellite annual-max daily rainfall, 334 grid nodes, 0.25° spacing | 100-yr RP depth (mm) per node          |
| Gauge-based (Indargarh) | 32-yr rain gauge record (1990–2021)                                | IDF table: mm/hr × 5 durations × 5 RPs |

**Five distributions fitted:** LP3, Gumbel EV1, Normal, Log-Normal, Pearson III

**Selection rule:** Highest 100-yr RP among distributions passing both chi-square and KS goodness-of-fit tests. Fallback to LP3 if none pass.

**Validation:** 13 tests, all passing. Tolerance 0.01 mm for individual distributions, 0.5 mm for selected RP.

**Three Excel bugs found and documented** (see Part 3).

Query CLI:

```
.venv/bin/python scripts/rainfall_query.py 29.0 88.5
```

Everything Else — Scaffolded, Not Built

| Module                                               | Status      |
|------------------------------------------------------|-------------|
| GIS — watershed delineation, basin geometry          | Placeholder |
| Hydrology — SCS CN runoff, Rational Method discharge | Placeholder |
| Hydraulics — Manning's HFL, afflux, freeboard        | Placeholder |
| Data pipeline — DEM, shapefile, raster ingestion     | Placeholder |
| API layer — FastAPI service                          | Placeholder |
| Report generation — Jinja2/WeasyPrint PDF            | Placeholder |

Part 3 — Excel Bugs Found During Development

Three discrepancies were found between the engine and the Excel workbooks during Test-Driven Development. Two are confirmed Excel bugs. One was our bug, now fixed.

### Bug 1 — Gumbel STDEVP (1 point, ~2mm impact)

Column 1 of the Gumbel sheet uses `STDEVP()` (population std, ddof=0) instead of `STDEV()` (sample std, ddof=1). Affects point (27.875°N, 89.125°E) only. Impact: 164.1mm in Excel vs correct 166.2mm.

**Fix:** Change `STDEVP` → `STDEV` in row 24, col B of the Gumbel sheet.

### Bug 2 — Chi-Square FREQUENCY Row Shift (Most Consequential)

In every chi-square sheet, the `FREQUENCY()` formula in columns 2–334 has a +4 row shift from a copy-paste error. It references rows 15–34 instead of the correct 11–30, including 4 statistical summary values (mean, std, Kn, Cs) in the frequency count and dropping the first 4 data years.

**Impact:** Changes the selected design distribution at 17 northern grid points (lat > 29°N). RP differences of up to 36mm at some sites.

**Fix:** Every column in row 51 of every chi-sq sheet must use:

```
=FREQUENCY(<column>11:<column>30, $C$2:$C$6)
```

### Bug 3 — Wilson-Hilferty Polynomial (Our Bug, Fixed)

Our initial implementation used the 5-term Taylor polynomial for the  $K_T$  frequency factor. This diverges from the exact cube-root formula at  $|Cs| > 1.9$ . Excel uses the exact formula. We fixed it — now matches Excel to 8 decimal places across all 334 points.

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## Part 4 — The SME Situation

### Who She Is

**Sudesna Biswas** — Deputy General Manager, Head of Hydrology & Hydraulics, Larsen & Toubro EDRC, Mumbai.

- 21 years experience, 14 years in lead/head roles
- M.Plan (Environmental Planning), SPA New Delhi + B.E. Civil, Nagpur University
- Executive Leadership, IIM Calcutta

- Currently heading hydrology for Delhi Airport, Bangalore Airport, Navi Mumbai Airport, Mumbai-Nagpur Expressway, Mumbai-Ganga Expressway, Bullet Train, Dedicated Freight Corridors
- Software: HEC-RAS, HEC-HMS, Civil Storm, Watergems, GIS MapInfo, AutoCAD
- Awards: L&T DELTA (design + digital innovation), L&T Pi Gold (process innovation)

This is not a junior consultant. She is one of India's most senior active infrastructure hydrology engineers.

## The Risk

The rainfall engine is SME-independent — fully documented, tested, and self-contained. The **four pending engines are not**. Their Excel workbooks, engineering judgment, and validation ground truth have not been extracted.

If contact ends before extraction: the project is not dead, but the pending engines would need a replacement SME to re-derive methodology from published standards — slower and more expensive.

## The Window

Contact is tense but not closed. This is the single highest-priority action in the entire project right now. One or two focused sessions is all that is needed — not to continue the relationship, but to not leave institutional knowledge locked in someone else's head.

## What to Extract (Priority Order)

**Session 0 — Get the files:** - Hydraulics calculation workbook (Manning's HFL, afflux, backwater) - Hydrology calculation workbook (SCS CN, Rational Method, Tc) - GIS/watershed calculation templates - Standard DPR report template for bridge hydrology

**Session 1 — GIS and Hydrology judgment calls:** - DEM source (SRTM 30m / Cartosat 10m / ALOS 12.5m?) - Time of concentration formula (Kirpich / Bransby-Williams / SCS lag?) - SCS Curve Number table for Indian land cover and soil groups - Hydrological Soil Group assignment for Indian soils - AMC default (AMC-II or adjusted?) - Rational Method C values by land cover - CWC regional zone for MPSB catchments - Design storm return period by structure type

**Session 2 — Hydraulics and standards:** - Manning's n values for natural streams, engineered channels, culverts - Afflux limit under IRC:5-2015 - Freeboard above HFL for bridge soffit - HEC-RAS cross-section setup for bridge crossings - Scour depth method (Lacey's / IRC:78 / Grip's) - IS and IRC codes that govern: IS: 5477, IS:4890, IRC:5, IRC:6, IRC:78, IRC:SP:13 - MPSB project specifics — sites, rivers, catchment sizes, why satellite data

**Full checklist:** docs/sme-extraction-checklist.md and .pdf

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## | Part 5 — The Venture Studio Model

### Structure

This is not two simultaneous ventures. It is a **sequenced studio**:

|                    |   |                           |
|--------------------|---|---------------------------|
| Phase 1 (now)      | → | Phase 2 (after traction)  |
| pibuilder          |   | Cold chain logistics      |
| EPC hydrology SaaS |   | Operated by Nancy Narayan |

### Budget

₹3.5–5 crores allocated across the studio pipeline. Sufficient for Phase 1 (full pibuilder build + first customers) if spent without distraction.

### The \$100M Exit Path

A Python library is worth nothing. The same algorithms as a platform — automating EPC hydrology workflows done manually in Excel across thousands of bridge and road projects in India and Southeast Asia — is a different story.

**The market:** India's EPC infrastructure market exceeds \$150B. Every bridge, culvert, road, airport, and railway project requires a hydrology design submission. These are currently done by engineers running Excel sheets, one project at a time.

**The moat:** Validated algorithms (13 tests, Excel-proven), Indian regulatory compliance built-in (IS:5477, IRC:5-2015, CWC), satellite data integration, no gauge dependency.

**Likely acquirers at exit:** Bentley Systems, Autodesk Infrastructure, ESRI, or a large Indian EPC firm (L&T, Sterlite, Dilip Buildcon). Strategic acquirers pay 8–12x ARR for compliance-heavy, domain-specific infrastructure software.

**The ceiling is real — but the full stack must be working first.**

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## **| Part 6 — Nancy Narayan and the Cold Chain Venture**

### **Who She Is**

**Nancy Narayan** — Sales Manager, CMA CGM, Dubai (May 1999–present).

- 25 years at one of the world's top 3 container shipping lines
- Grew freight volume from 5,000–15,000 TEUs (2014) to ~100,000 TEUs (2024)
- Improved repeat business from 20% to 65%
- Deep network: freight agents, forwarders, container ops, rate negotiation
- Career: Logistics Executive → Customer Service Supervisor → Asst. Sales Manager → Sales Manager

### **Why This Is an Asset, Not a Subsidy Play**

The 3% NHB loans and government cold chain subsidies are attractive. They are also available to every competitor. Nancy's 25-year CMA CGM network — agent relationships, freight pricing intelligence, port ops knowledge — is not.

That is the moat. Build around her, not around the subsidy.

### **Open Questions Before Committing**

- Is she planning to leave CMA CGM, and on what timeline?
- Will she operate from India or remotely from Dubai?
- Does she have India-specific cold storage and freight contacts, or primarily UAE/international?
- Written agreement on roles, equity, decision rights, and exit terms — before the first rupee moves. A 25-year career means strong opinions. That is an asset and a potential collision point. Document it early.

### **Sequencing**

Cold chain does not start until pibuilder has traction — paying customers and proven unit economics. Any earlier and the studio model collapses into two underfunded, distracted ventures.

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## | Part 7 — Staged Project Plan

### Stage 1 — Core Rainfall Engine ✓ Done

Rainfall frequency analysis, IDF tables, CLI query tool, SME spec PDF. Validated against MPSB + Indargarh Excel. 13 tests passing.

### Stage 2 — Complete the Calculation Stack (3-4 months)

Implement remaining engines in dependency order, each built TDD against Excel ground truth extracted by Sudesna **before** coding begins:

1. GIS engine — watershed delineation (WhiteboxTools D8), basin area, slope, longest flow path
2. Hydrology engine — SCS CN runoff depth, Rational Method + SCS peak discharge
3. Hydraulics engine — Manning's HFL, afflux, freeboard (IRC:5-2015)
4. Data pipeline — DEM, shapefile, rainfall raster ingestion and ETL

**Gate:** SME extraction must complete before Stage 2 starts.

### Stage 3 — Automated Report Generation (1-2 months)

Jinja2/WeasyPrint PDF reports: design basis, distribution selection, HFL tables, freeboard calculations. This is what an engineer submits to a client. Automating it is the first visible, billable time saving.

### Stage 4 — API and Multi-Project Platform (2-3 months)

FastAPI service layer. Move from single-project scripts to a platform that runs analyses for many bridge/culvert sites in batch. Add lightweight web UI or REST API. This is where it becomes a product, not a library.

### Stage 5 — First Paying Customers (3-6 months)

- SaaS pricing per project or per seat
- 3-5 EPC pilot customers (Indian market first — regulatory alignment built in)
- Audit trail and version control for design values (regulatory requirement)
- ISO/BIS compliance documentation

## Stage 6 — Scale or Exit (12-24 months post Stage 5)

Raise a Series A on ARR, or approach strategic acquirers. The asset at this point: validated algorithms, regulatory compliance, paying EPC customers, and a defensible moat.

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## Part 8 — Key Risks and Mitigations

| Risk                                   | Severity | Mitigation                                                      |
|----------------------------------------|----------|-----------------------------------------------------------------|
| SME contact ends before extraction     | Critical | Run extraction sessions immediately — this week                 |
| Budget split across ventures too early | High     | Cold chain starts only after pibuilder traction                 |
| Nancy not available / not relocating   | Medium   | Confirm availability and terms before building plans around her |
| No paying customers validate the model | High     | Stage 5 gate — get 3 paying customers before scaling            |
| Regulatory changes to IRC/IS codes     | Low      | Engine is parameterised — update constants, not architecture    |

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## Part 9 — Immediate Next Actions

1. **This week:** Contact Sudesna. Frame it professionally and narrowly — "one or two sessions to transfer project files before we wrap up." Not a relationship repair. A knowledge rescue.
  2. **Get the files first** (Priority 0 in checklist) — hydraulics, hydrology, watershed workbooks. Everything else can be reconstructed from published standards. Her Excel judgments cannot.
  3. **Confirm Nancy's availability and timeline** before the cold chain venture features in any investor conversation.
  4. **Start Stage 2** only after the extraction checklist is substantially complete.
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## Appendix — Key References

| Reference                                                     | Used for                                                |
|---------------------------------------------------------------|---------------------------------------------------------|
| Wilson & Hilferty (1931). PNAS 17(12).                        | Exact K_T formula                                       |
| Gumbel, E.J. (1958). Statistics of Extremes.                  | Gumbel EV1                                              |
| Chow, Maidment & Mays (1988). Applied Hydrology. McGraw-Hill. | General frequency analysis                              |
| IS:5477 Part 4 (1971). Bureau of Indian Standards.            | LP3 as Indian hydrology standard                        |
| CWC (2002). Flood Estimation — Western Himalayas Zone 7.      | Duration scaling exponent 1/3                           |
| IRC:5-2015.                                                   | Standard Specifications for Road Bridges                |
| IRC:SP:13.                                                    | Guidelines for the Design of Small Bridges and Culverts |